

Masoud Ardeshiri

Ph.D. In Electrical Engineering Power Electronics -Electric Drives

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Summary

Result-driven electrical engineer with expertise in power systems, power electronics, energy storage, and electric drives. Currently, I am pursuing a Ph.D. at the Warsaw University of Technology, specializing in the integration of renewable energy (PV-Wind Turbine), battery energy storage systems (BESS), and advanced energy management. Skilled in designing, modeling, and optimizing electrical systems for high efficiency and reliability. Known for strong problem-solving abilities, collaborative teamwork, and adaptability in complex engineering environments.

Work Experience

Junior Electrical Engineering

Oil Refinery Company Shiraz-Fars-Iran 2019-2020

- Designed and implemented internal cabling and wiring for boiler and water units.
- Conducted maintenance and troubleshooting of internal electrical grids, ensuring uninterrupted operations.
- Collaborated with engineering teams to optimize system reliability and safety.

Junior Electrical Engineering

Petrochemical Company Shiraz-Fars-Iran 2013-2014

- Assisted in wiring/cabling for main complex electrical installations and control rooms.
- Reviewed and interpreted electrical schematics for installation and commissioning activities.
- Supported troubleshooting operations to minimize downtime and enhance efficiency

Electrical Engineer Intern

Central Electricity Authority Shiraz-Fars-Iran 2011-2012

- Operated and maintained High and Medium Voltage substations.
- Performed detailed analysis of power consumption patterns for performance optimization.
- Participated in overhaul and maintenance projects, gaining hands-on experience with DigSilent simulation tools.

Electrical Engineer Intern

Oil Refinery Company Shiraz-Fars-Iran 2006-2007

- Assisted in installation, commissioning, and maintenance of internal electrical networks.
- Familiarized with Gas Turbine Unit (GTU) control board wiring and functionality.
- Maintained and updated operational records and technical documentation

Technical Skills

- Control, Analysis of Power Converters such as AC-DC/DC-AC /DC-DC, Buck/Boost/ Buck-Boost
- Renewable Energy Systems
- Photovoltaic Systems (PVs) Analysis
- Battery Storage Systems, Control and Analysis
- Energy Management, Analyzing/ Controlling/Monitoring/Trouble Shooting
- Electric Drives, Synchronous and Asynchronous Generator system Simulation and Analysis
- AC / DC Electrical Systems / Network Caballing / Wiring Simulation and Implementation
- Testing, Commissioning and Computer Modeling
- Design, Analysis, Monitoring and Trouble shooting
- Power Systems Analysis
- Power Quality
- Wind Turbine Modeling and Analysis (Off-Grid/Connected Grid)

Language

English Full Professional Proficiency

Persian Native/Bilingual



Education

Ph.D. Electrical Engineering

Warsaw University of Technology  **2021-2026**

- *Research Focus: Renewable Energy (Wind Turbine, PV Systems), Energy Management, Control of Power Electronics Converters, Electric Drives.*
- *Thesis: Analysis of Power Generation Systems Supported by Hybrid Battery-Supercapacitor Energy Storage Systems.*

M.Sc. Electrical Engineering

Azad University of Kazeroon Branch-Iran  **2015-2017**

- *Research Focus: Renewable Energy, Power Electronics, Control, Electric Drives*
- *Thesis: Active and Reactive Power Control in Wind Turbines Based on DFIG Using Type-2 Fuzzy Logic Controller Tuned by PSO Algorithm.*

B.Sc. Electrical Engineering

Azad University of Shiraz Branch-Iran  **2011-2013**

- *Research Focus: Power Systems-Analysis of the power systems*
- *Thesis Topic: Troubleshooting Power Transformer by Using Oil-Soluble Gas Analysis*

Software Skills

- MATLAB Simulink
- PISIM
- PLCES (Plexim)
- P-Spice
- Electronic Work Bench
- Arduino
- DigSilent
- Latex
- Microsoft Office (Excell)
- Microsoft Visio

General Skills

- Problem Solving
- Research and Development
- Technical presentation
- Project Reporting, management, Scheduling
- Team Working- Goal Oriented Mindset

Interests

- Modern A-I Technologies
- Aviation/Aerospace
- Learning new languages
- Geography